

SHIVA KUMAR HAZARI

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Field IoT Engineer | B.Tech (IoT) & Diploma (ECE)



PROFILE SUMMARY

Industrial IoT & Edge Systems Engineer with hands-on experience architecting and deploying end-to-end telemetry pipelines across Amazon GRE F logistics sites. Experienced in bridging physical instrumentation (RS-485 Modbus, LoRaWAN IN865, Pulse Meters) with enterprise cloud platforms (AWS IoT Core, X.509 mTLS). Proven track record of engineering full-stack Python automation tools that slashed gateway commissioning time by 95% (15 min → 4 min) while maintaining 99%+ data reliability. Google open-source contributor with a strong foundation in embedded C/C++, Linux edge gateways, and real-time SCADA HMI systems.

EDUCATION

Bachelor of Technology – IoT Malla Reddy College of Engineering and Technology, Hyderabad, Telangana, India	Oct 2022-Apr 2025 CGPA: 7.2
State Board of Technical Education and Training- ECE ST.Mary's Engineering College, Hyderabad, Telangana, India.	Jun 2018-May 2021 CGPA: 6.98
SSC (10th Grade) Z.P.H.S Haveli Ghanapur , Medak, India.	Jun 2017-May 2018 GPA: 8.2

TECHNICAL SKILLS

Embedded & Edge Hardware:	ESP32 (Tensilica Xtensa), Raspberry Pi 3/4/5, MultiTech LoRaWAN Gateways (MTCAP3), RS-485 Modbus Transceivers, Pulse Counters / Utility Meters, Sensors & Actuators, Circuit Analysis.
IoT & Industrial Protocols	: MQTT, MQTT over WebSockets, LoRaWAN (IN865 / Basic Station / CUPS), RS-485 Modbus RTU, CoAP ,TCP/IP Sockets, HTTP/REST, 4G LTE Backhaul, Wi-Fi (802.11 b/g/n).
Cloud & Edge Security	: AWS IoT Core (ATS Endpoints, Device Shadow, Rules Engine), Mutual TLS (mTLS / X.509 PKI Certs) , Mosquitto MQTT Broker, IoT Policy Management, Zero-Touch Device Provisioning.
Programming & Scripting	: C, C++ (Embedded / ArduinoJson / PubSubClient), Python 3 (Asyncio, Paho-MQTT, AWS SDK v2 FastAPI), Java, ES6+ JavaScript (HTML5 Canvas 2D, WebSockets), SQL, Bash Scripting.
Development, Automation & QA:	Git, GitHub Actions (CI/CD), Maven, JUnit 5 / JUnit 4 (Google OSS Contributor), Selenium WebDriver, Postman, Linux / Raspberry Pi OS, VS Code, PlatformIO.
Methodologies & Architecture:	Edge-to-Cloud Architecture, SCADA / Web HMI, Agile / Scrum, SDLC, Reverse Engineering, RF & Signal →_→ Interference Analysis.

EXPERIENCE

Milvian group Hyderabad / Pan-India	Jan 2026 – Present
- Engineered Python automation scripts to streamline MultiTech Gateway provisioning and configuration, reducing manual setup time from 15 minutes down to 3–4 minutes per unit across large-scale deployments. - Researched and analyzed legacy utility meters on-site, reverse-engineering technical specifications to extract pulse outputs and successfully integrate non-smart hardware into modern IoT tracking systems. Integrated electrical meters via RS-485 (Modbus) and deployed Milesight Pulse Counters for high-precision utility monitoring. Architected and deployed robust telemetry data pipelines using LoRaWAN and LTE backhaul to ensure seamless data flow from edge sensors to cloud platforms. - Conducted site-specific signal analysis and LoRaWAN tuning to mitigate structural interference in complex warehouse environments.	
Google Remote	Apr 2025 - Present
Open Source Contributor – JUnit 5 Migration - Selected to contribute to Google's high-impact fhir-data-pipes open-source project on GitHub under Issue #1103. - Migrating legacy unit test modules from JUnit 4 to JUnit 5, ensuring modern standards and compatibility. - Actively collaborating with Google engineers through modular PRs and peer-reviewed code contributions. - Participating in bi-weekly developer syncs to align with Google's project scope and timelines. - Utilizing Maven, GitHub Actions, and CI workflows to validate and streamline test suite migration.	

PROJECTS

Nexus Core: Industrial IoT Edge-to-Cloud Platform AWS IoT Core, Raspberry Pi, ESP32 (C++), Python, MQTT, X.509 mTLS, JavaScript, Chart.js Link	Jun 2026 – Aug 2026
- Architected an End-to-End 3-Tier Industrial IoT (IIoT) Ecosystem comprising ESP32 smart edge nodes, a Raspberry Pi Edge Gateway, AWS IoT Core, and a real-time Web SCADA dashboard, delivering sub-50ms telemetry streaming and continuous machine health monitoring across multiple industrial nodes. - Engineered Robust C++ Embedded Firmware on ESP32 utilizing ArduinoJson and PubSubClient to sample 8+ electrical and physical telemetry parameters (Active Power, Voltage, Current, Flow Rate, Line Pressure, Cumulative kWh); optimized packet memory by configuring static 512-byte payload buffers and engineered an eFuse MAC-derived client ID system with automated Wi-Fi/MQTT reconnection state machines. - Developed a High-Performance Python Edge Gateway on Linux/Raspberry Pi utilizing AWS IoT Device SDK v2 and Paho-	

MQTT in a dual-broker architecture; ingested local MQTT streams, enriched payloads with gateway-level epoch timestamps, and securely relayed telemetry to AWS IoT Core endpoints over mutual TLS (mTLS / X.509 certificates) with QoS 1 reliability.

- Constructed "Nexus Core" Real-Time Web HMI & Diagnostics Engine in modular ES6+ JavaScript, integrating direct MQTT-over-WebSockets with automatic failover to simulated streams; designed a zero-dependency HTML5 Canvas time-series charting engine supporting High-DPI retina rendering, anomaly threshold highlighting, dynamic time ranges (1H/24H/7D), and an interactive alert lifecycle manager (OPTIMAL → CRITICAL → RESOLVED).

Automated LoRaWAN Gateway Provisioning & Telemetry Platform | Python, FastAPI, LoRaWAN

July 2026 – Aug 2026

, Selenium, REST APIs | [Link](#)

- Architected and developed a full-stack IoT provisioning platform using Python, FastAPI, and JavaScript, automating the end-to-end commissioning of MultiTech LoRaWAN gateways (MTCAP3-LEU7) and slashing setup time from 15 minutes to ~45 seconds (95% efficiency gain).
- Engineered an asynchronous 11-stage hardware orchestration engine, interfacing with edge REST APIs to automate LoRaWAN IN865 channel plans, 4G LTE cellular APN configurations, SNTP time synchronization, and scheduled maintenance reboot cycles.
- Implemented automated cloud-to-edge TLS certificate injection, utilizing Selenium WebDriver to automate cloud portal (Atria AWS LoRaWAN) device registration and dynamically link .trust, .crt, and .key cryptographic pairs into CUPS Basic Station interfaces.
- Built real-time telemetry streaming over Server-Sent Events (SSE) and thread-safe queue dispatchers, delivering live gateway execution status, connection heartbeats, and console logs to a responsive web dashboard.
- Designed low-level network discovery algorithms, leveraging socket probing (TCP port 443) and ARP table parsing to dynamically resolve device MAC addresses and IP bindings with zero manual input.
- Developed dual-persistence audit logging and data pipelines using OpenPyXL and JSON, synchronizing gateway metadata (Serial, Dev EUI, MAC, UUID, IMEI) for inventory tracking and 1-click fleet export.

CERTIFICATIONS

Agile Scrum in practice from Infosys springboard.
Programming using Java from Infosys springboard
Data Structures and Algorithms using Java

Database Management System from Infosys springboard
SQL(Basic) certificate from hackerrank.
Git&Github from Infosys springboard.

ACHIEVEMENTS

- Participated in national level Hackathon organized by IIT Bhilai.
 - Awarded a top 20 ranking on GeeksforGeeks(College level).
 - Achieved a Gold badge on HackerRank by solving complex Java challenges, ranking in the top 3.4% out of 2.2 million users.
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